Marketing Analytics

Marketing Decisions and AI-Assisted Coding

By Nathan Yang

Session 01



A Renewal Offer

- A subscription firm can contact 5,000 eligible customers.
- The decision concerns incremental contribution over the next 30 days.
- Past spending, consent, and account history are available before contact.

Analytical Questions

- Description: how often do customers renew?
- Prediction: who is likely to renew?
- Causal effect: how does an offer change renewal?
- Decision: which contacts generate positive incremental contribution?

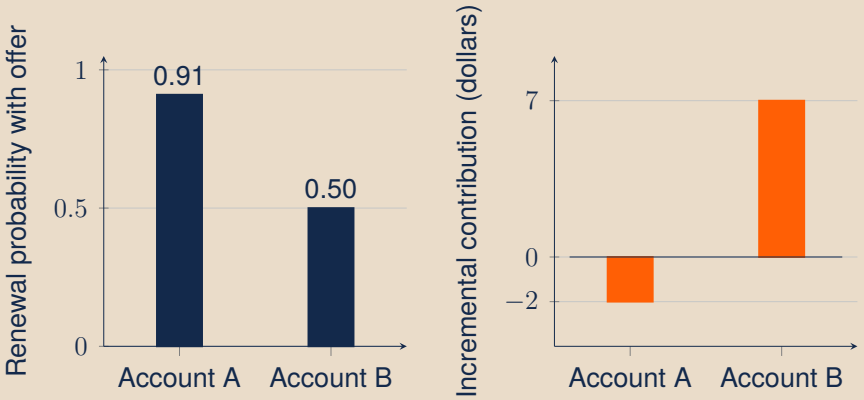
Incremental Contribution

For account i, compare renewal under an offer and no offer.

$$\Delta\pi_i = m_i\{p_i(1) - p_i(0)\} - c_i$$

Contribution m_i and contact cost c_i are dollars per account over the stated horizon.

Response and Incremental Value



Synthetic accounts. Contribution is 100 dollars and contact cost is 3 dollars.

The Observation Unit

- A product line, order, customer, and customer-month are different units.
- The key identifies one row at the intended level.
- Aggregation changes the weights used in an average.

Python Objects and Order Totals

Three synthetic product lines belong to two orders.

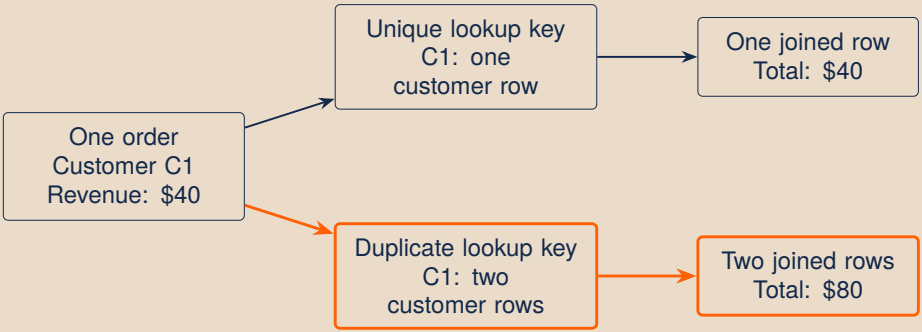
```
lines = [('A', 2, 10), ('A', 1, 5), ('B', 1, 12)]
orders = {}
for order, quantity, price in lines:
    orders[order] = orders.get(order, 0) + quantity * price
total = sum(orders.values())
aov = total / len(orders)
assert (total, aov) == (37, 18.5)
```

Check: total sales are 37 dollars and average order value is 18.50 dollars.

Functions and Data Contracts

- Inputs have names, types, units, and allowed missing values.
- A function returns a defined output and an exception report.
- Returns, canceled orders, and duplicate lines need explicit rules.

A Duplicate Lookup Doubles Sales



A valid-looking join can double revenue.
Check lookup-key uniqueness before merging, then reconcile totals.

Synthetic join. One 40-dollar order matches two customer records.

An AI Coding Request

- Return one row per order and an exceptions table.
- Treat negative quantities as returns under the stated rule.
- Verify the 37-dollar example and reject missing order IDs.
- Explain each transformation before adding the next step.

Code Errors and Missing Evidence

- A repeated daily total requires a correction to aggregation.
- Unrecorded food requires additional measurement or a recorded-behavior target.
- A coupon decision needs evidence about untreated outcomes.

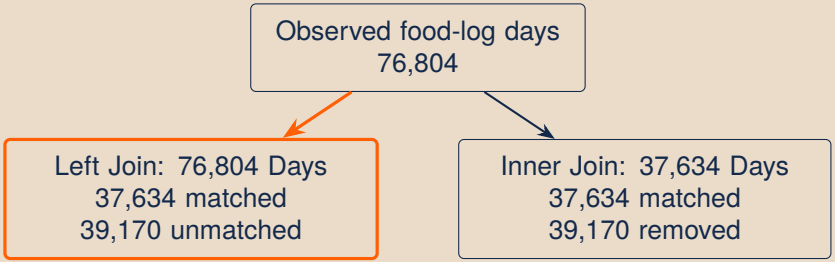
Verification With Counterexamples

- A duplicate lookup must trigger a failed key check.
- An unmatched customer must remain visible in the assessment.
- An empty input needs an explicit result.
- Order totals must reconcile with eligible product lines.

mHealth Records

- Food entries describe recorded items on observed user-days.
- The separate daily table covers only part of the food-log history.
- Reported daily calories repeat across item rows.

Join Choice Changes the Sample



Same source keys; different analysis populations.
An unmatched day is still an observed food-log day.

Released mHealth cohort: 76,804 observed user-days.

Coding Exercise: An Order Assessment

- Implement aggregation with a coding assistant.
- Add a return, a missing identifier, and a duplicated lookup key.
- Report how each change affects rows, totals, and exceptions.

A Reproducible Analysis

- Relative paths identify inputs and outputs.
- A fresh run reconstructs results without hidden notebook state.
- Record package versions and consequential AI contributions.

A Campaign Specification

- Define the eligible customers and contact date.
- State the contribution horizon and relevant costs.
- Identify the evidence needed for the untreated outcome.
- Choose a numerical check that would reveal a wrong denominator.

Customer Data in an AI Workflow

- Use only data authorized for the tool and task.
- Minimize fields and use synthetic examples when they suffice.
- Check where uploads, logs, and generated files are stored.

Decision Discussion

- Which customer receives the renewal offer?
- What changes if every treated renewal receives a discount?
- Which future variable would invalidate a Monday-morning forecast?